First, we will derive an equation for the gain at zero frequency. The idea is to force each of the input samples to have a value of *one*, resulting in each of the output samples having a value of *G*, the gain of the system we are trying to find. We will start by writing the recursion equation, the mathematical relationship between the input and output signals:

$$y[n] = a_0x[n] + a_1x[n-1] + a_2x[n-2] + \cdots + b_1y[n-1] + b_2y[n-2] + b_3y[n-3] + \cdots$$

Next, we plug in *one* for each input sample, and *G* for each output sample. In other words, we force the system to operate at zero frequency. The equation becomes:

$$G = a_0 + a_1 + a_2 + a_3 + \cdots + b_1G + b_2G + b_3G + b_4G \cdots$$

Solving for *G* provides the gain of the system at zero frequency, based on its recursion coefficients:

EQUATION 33-7
DC gain of recursive filters. This relation provides the DC gain from the recursion coefficients.

$$G = \frac{a_0 + a_1 + a_2 + a_3 \cdots}{1 - (b_1 + b_2 + b_3 \cdots)}$$

To make a filter have a gain of *one* at DC, calculate the existing gain by using this relation, and then divide all the "a" coefficients by *G*.

The gain at a frequency of 0.5 is found in a similar way: we force the input and output signals to operate at this frequency, and see how the system responds. At a frequency of 0.5, the samples in the input signal alternate between -1 and 1. That is, successive samples are: 1, -1, 1, -1, 1, -1, 1, etc. The corresponding output signal also alternates in sign, with an amplitude equal to the gain of the system: *G*, -*G*, *G*, -*G*, *G*, -*G*, etc. Plugging these signals into the recursion equation:

$$G = a_0 - a_1 + a_2 - a_3 + \cdots - b_1G + b_2G - b_3G + b_4G \cdots$$

Solving for *G* provides the gain of the system at a frequency of 0.5, using its recursion coefficients:

EQUATION 33-8
Gain at maximum frequency. This relation gives the recursive filter's gain at a frequency of 0.5, based on the system's recursion coefficients.

$$G = \frac{a_0 - a_1 + a_2 - a_3 + a_4 \cdots}{1 - (-b_1 + b_2 - b_3 + b_4 \cdots)}$$